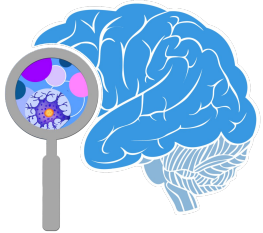


Anaerobic respiration

Lesson objectives:

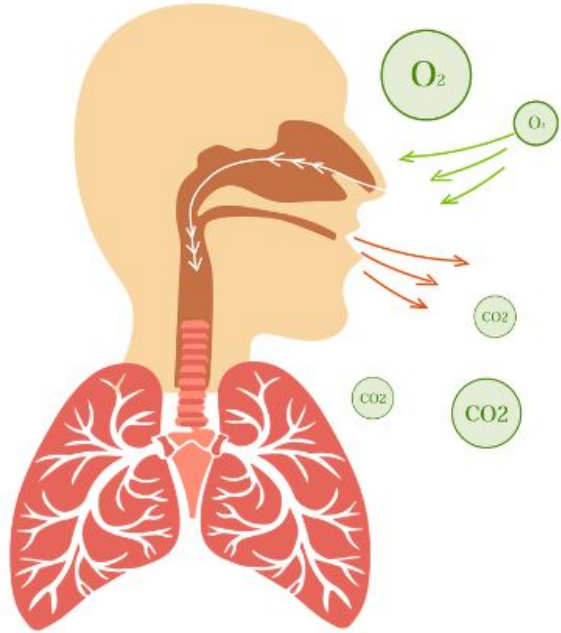
- To state the word equation for anaerobic respiration.
- To describe differences between aerobic and anaerobic respiration.



Respiration is often confused with breathing - But this is not the case.

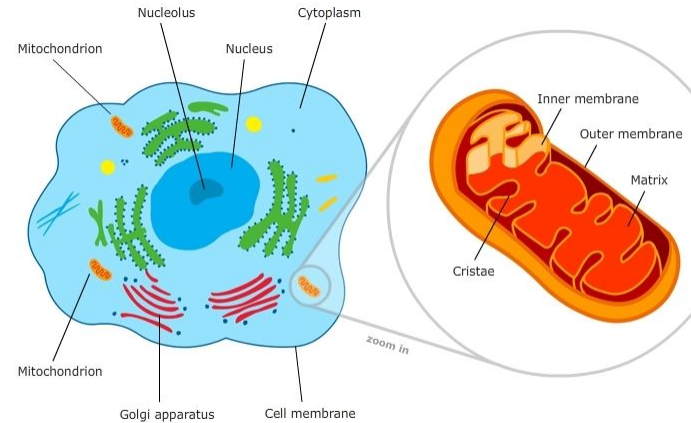
What is respiration? You have *two minutes* to answer this question in your books.

Breathing vs Respiration



Breathing is the physical process of inhaling and exhaling

VS



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Respiration is the biochemical process of energy production in cells.

Anaerobic respiration

If aerobic respiration uses oxygen to break down glucose for energy, but anaerobic does not require oxygen - **when do you think we might use anaerobic respiration?**

- When exercising **intensely for short periods of time**, the body cannot easily get the oxygen it needs.
- Paralympian Jonnie Peacock respire anaerobically during the 100m sprint.



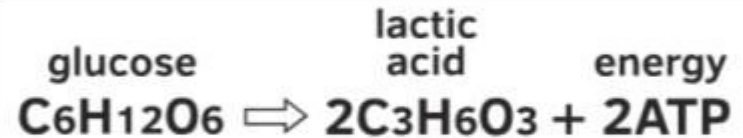
How does it work?

Intense exercise requires a large oxygen supply to the muscles. This will release the energy that the muscles need to work hard.

- The body cannot provide oxygen quickly enough so glucose has to be broken down and turned into energy in another way.

Anaerobic respiration:
glucose $\rightarrow$ lactic acid

Equation: Key point to know

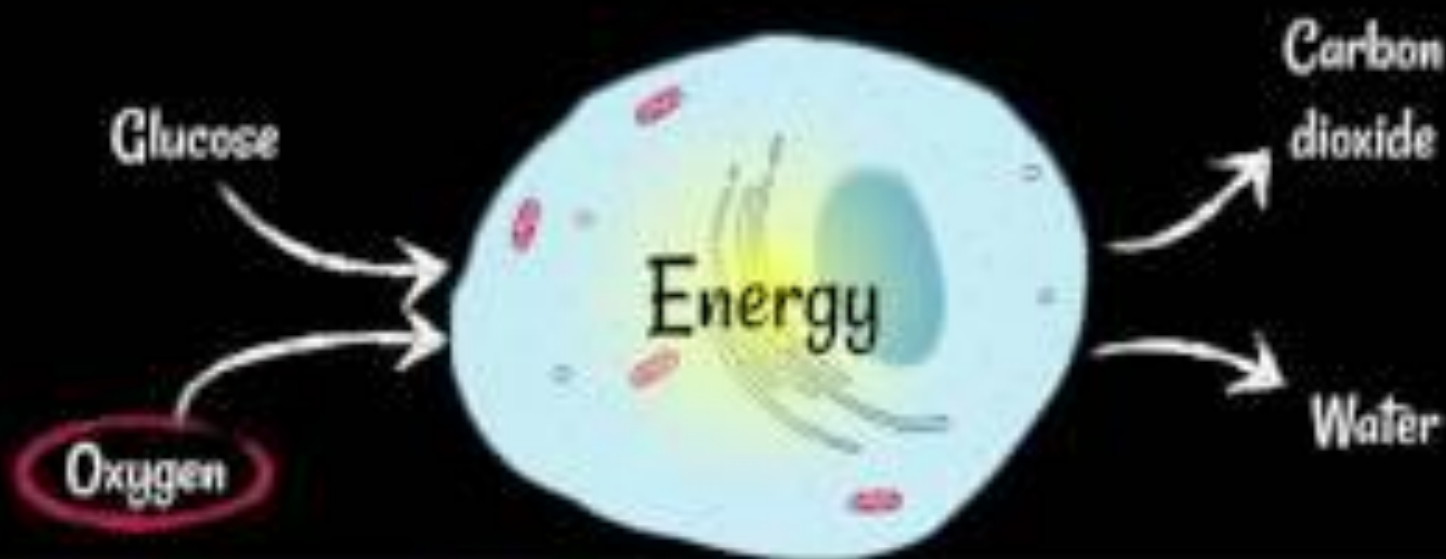


Effects of intense exercise

Oxygen breaks lactic acid down into carbon dioxide and water but without oxygen, lactic acid is able to build up.

When the body respire anaerobically, there is a build-up of lactic acid, which causes pain in the muscles.

Less energy is released during anaerobic respiration than aerobic respiration, so it cannot be sustained for prolonged periods of time.



Aerobic respiration

Glucose + Oxygen $\longrightarrow$ Carbon Dioxide + Water

Aerobic vs Anaerobic respiration

	Aerobic respiration	Anaerobic respiration
Oxygen		
Glucose breakdown		
End product(s)		
Energy released		

Aerobic vs Anaerobic respiration

	Aerobic respiration	Anaerobic respiration
Oxygen	Needed	Not needed
Glucose breakdown	Complete	Incomplete
End product(s)	Carbon dioxide and water	Lactic acid
Energy released	Large amounts	Small amounts

Aerobic vs Anaerobic respiration

...write down the similarities and differences between aerobic and anaerobic respiration.



Keywords: oxygen, glucose, carbon dioxide, water, lactic acid, ATP, energy

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